

TOOLS & TECHNOLOGIES

Python / Pandas

MySQL / SQL

Power BI

ANALYTICS CASE STUDY

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

3,900

18

4

Categories

Strategic Engagement Overview

DATASET

3900 records purchase records across 18 features including demographics, spend behavior, and shipping preferences

OBJECTIVE

Identify spending patterns, segment customers, and surface actionable product and revenue insights for strategic decisions

METHODOLOGY

End to end Pipeline: Python data preparation → MySQL SQL analysis → Power BI interactive dashboard visualization

KEY FINDING

Male customers drive 67.8% of total revenue; Loyal segment represents 80% of the customer base at 3,116 customers

Strategic Challenge

How can shopping behavior data be transformed into actionable business intelligence?

01

Revenue Attribution

Which customer segments generate the most revenue?

02

Demographic Influence

How do demographics influence purchasing behavior?

03

Product Performance

Which products and categories lead in performance?

Data Foundation

3,900

18

4

Product Categories

\$20 - \$100
Purchase Range

Customer Demographics

- › Age groups & cohorts
- › Gender & location
- › Subscription status

Purchase Details

- › Item & category
- › Purchase amount (USD)
- › Season, size, color

Shopping Behavior

- › Discounts & promo codes
- › Frequency & reviews
- › Shipping type

Three-Phase Analytical Pipeline

01

Python Data Preparation

- Load dataset with pandas
- Explore structure & statistics
- Impute missing review ratings
- Rename columns to snake_case
- Engineer age groups & frequency
- Export cleaned data to MySQL

02

SQL Business Analysis

- Revenue by gender segment
- Top 3 products per category
- Express vs Standard delivery
- Subscription & repeat buyer analysis
- Customer segmentation by frequency
- Age group revenue attribution

03

PowerBI Dashboard

- Import cleaned MySQL dataset
- Build interactive KPI cards
- Revenue by age group charts
- Category sales breakdown
- Subscription status visuals
- Drill-down filter panel

01

SECTION

Python Data Preparation

Building the analytical foundation

Data Preparation Process

01

Data Loading

Imported dataset using pandas for analysis

02

Initial Exploration

Checked structure with `df.info()` and summary statistics with `.describe()`

03

Missing Data Handling

Imputed missing review ratings using median by product category

04

Standardization

Renamed columns to snake_case for better readability

05

Feature Engineering

Created age groups and purchase frequency days from raw data

06

Database Integration

Connected to MySQL and loaded cleaned DataFrame

02

SECTION

SQL Business Analysis

Revenue intelligence through structured query analysis

SQL Analysis: Revenue & Subscription

```
SELECT gender, SUM(purchase_amount) AS total_revenue  
FROM shopping_data GROUP BY gender
```

MALE

\$157,890

67.8% of total revenue

FEMALE

\$75,191

32.2% of total revenue

KEY INSIGHT • Male customers outspend Female 2:1 — (\$157,890 vs \$75,191). Gender-targeted campaigns represent the highest-impact revenue lever.

02 • AVG SPEND VS TOTAL SPEND (SUBSCRIPTION)

```
SELECT subscription_status,  
       COUNT(*) AS total_customer,  
       AVG(purchase_amount) AS avg_spend,  
       SUM(purchase_amount) AS total_spend  
FROM shopping_data GROUP BY subscription_status
```

SUBSCRIBER

1,053 customers

\$59.49 avg | \$62,645 total

NON-SUBSCRIBER

2,847 customers

\$59.87 avg | \$170,436 total

KEY INSIGHT • Subscribers and non-subscribers show nearly identical avg spend (\$59.49 vs \$59.87) — subscription drives loyalty, not spend uplift.

SQL Analysis: Product & Delivery

03 · TOP 3 PRODUCTS FROM EACH CATEGORY

RANK() OVER (PARTITION BY category ...)

ACCESSORIES

- #1 **Jewelry**
171 orders
- #2 **Sunglasses**
161 orders
- #3 **Handbag**
148 orders

CLOTHING

- #1 **Blouse**
172 orders
- #2 **Pants**
171 orders
- #3 **Shirt**
169 orders

STANDARD FOOTWEAR

\$60.48 Sandals

- #1
160 orders
- #2
150 orders

STANDARD OUTERWEAR

\$58.46 for Jacket

- #1
163 orders
- #2 **Coat**
161 orders

04 · EXPRESS VS STANDARD DELIVERY

Express customers avg \$2.02 more than Standard — logistics tier has a measur

SQL Analysis: Customer Segmentation

```
SELECT CASE  
  WHEN frequency >= 50 THEN 'LOYAL'  
  WHEN frequency >= 1 THEN 'RETURNING'  
  ELSE 'NEW'
```

LOYAL

3,116

80.0% of customers

RETURNING

701

18.0% of customers

NEW

83

2.1% of customers

BUSINESS IMPACT • LOYAL customers (3,116) form the dominant segment — targeted retention programs for this group deliver maximum ROI

06 • AGE GROUP REVENUE ATTRIBUTION



Young Adults lead total purchasing at \$62,143 — Middle-Aged follows closely at \$59,197

SQL Analysis: Product Performance

```
SELECT item_purchased, ROUND(AVG(review_rating), 2) AS avg_rating
FROM shopping_data
GROUP BY item_purchased ORDER BY avg_rating DESC LIMIT 5
```

#1	Gloves	3.86	
#2	Sandals	3.84	
#3	Boots	3.82	
#4	Hat	3.8	
#5	Handbag	3.78	

All top 5 items cluster tightly between 3.78–3.86 — indicating consistently high customer satisfaction

01★

Top Review Products

Identified top5productswithhighest average review ratings for quality benchmarking

02

Category Leaders

Determined top 3 products from each category for inventory planning and stock optimization

03 STRATEGIC OPPURTUNITY

Cross reference top rated items with category leaders to prioritize high performing , high satisfaction for promotional focus

Critical Findings & Strategic Implications

\$233,081

Across 3,900 purchase transactions

67.8%

\$157,890 vs \$75,191 Female

80%

Customer Base: Loyal Segment

3,116 of 3,900 customers

Revenue by Gender

Revenue attribution by gender segment guides targeted marketing investment and channel allocation decisions

Discount Strategy Impact

Customers using discounts but spending above average represent a margin optimization opportunity — precision discount targeting

Subscription Behavior

Subscribers (1,053) and non-subscribers show nearly identical avg spend (\$59.49 vs \$59.87) — subscription drives loyalty, not spend uplift

Repeat Buyer Link

Examined if repeat buyers are more likely to subscribe — enables predictive subscription upsell campaign design

03

SECTION

Power BI Dashboard

Interactive executive analytics visualization

Interactive Analytics Platform

Customer Behaviour Dashboard - Power BI Interactive

\$233K

Total Revenue

3,900

Transactions

\$59.7

Avg Order Value

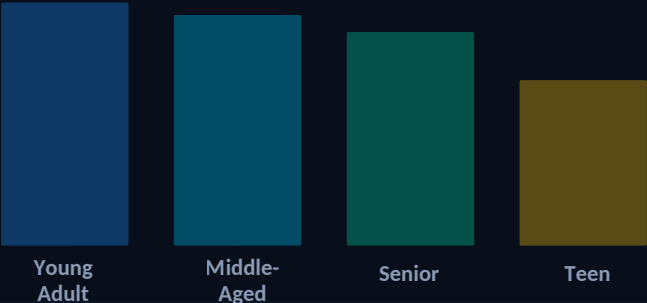
80%

Loyal Segment

1,053

Subscribers

Revenue by Age Group



Category Split

Accessories 29%

Clothing 28%

Footwear 25%

Outerwear 18%

Subscription

27%

Subscribed

Interactive Visuals

Dynamic charts for multi-angle exploration

Drill-Down Filters

Filter by category, season, location, segment

Strategic Insights

Revenue trends and product performance at a glance

Six Action Priorities

R1

Gender-Targeted Marketing

Male customers = 67.8% revenue. Develop female-targeted campaigns to balance distribution and expand addressable market.

R2

Subscription Upsell Program

73% non-subscribed yet spending equally (\$59.87 avg). Value-added subscription tier with exclusive benefits could drive conversion without price resistance.

R3

Inventory Optimization

Category leaders (Jewelry, Blouse, Jacket) and top-rated items (Gloves, Sandals, Boots) should guide procurement priorities and promotional calendar.

R4

Loyal Segment Retention

80% of customers are LOYAL (3,116). Loyalty rewards and personalized offers protect the most revenue-critical customer base.

R5

Discount Strategy Refinement

Customers using discounts but spending above average = optimization opportunity. Tune discount thresholds to maximize margin without losing volume.

R6

Age Group Targeting

Young Adults lead at \$62,143 total spend. Tailored product lines and digital marketing channels for this segment can amplify top-spending behavior.

Strategic Decisions Enabled

Strategic Decisions

Revenue by gender, discount strategy impact, and subscription spending patterns guide marketing and retention strategies

Product Insights

Top review products and category leaders inform inventory planning and merchandising decisions

Customer Segments

Repeat buyer subscription likelihood and segment revenue contribution enable targeted campaigns

IMPACT SUMMARY

3,900

SQL Queries

8

Categories

4

Segments

3

\$233K

Python / Pandas

MySQL / SQL

Power BI

ANALYTICS CASE STUDY

Customer Shopping Behavior Analysis

End-to-end analytics engagement covering data engineering, SQL analysis, and executive BI visualization.

3,900 purchase records analyzed across 18 features

8 SQL business queries with structured result sets

4 product categories benchmarked for performance

3 customer segments identified with revenue attribution

Interactive Power BI dashboard built for stakeholders